A TWO-STAGE SUMMARIZATION PIPELINE FOR NEWS ARTICLES:
EXTRACTION-GUIDED ABSTRACTIVE GENERATION

AIMLCZG628T: DISSERTATION – MID-SEMESTER REPORT

by

PUTHINEEDI VENKATA SAI CHARAN
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Under the Supervision of

Veeraswamy Ponnuru
Lead Quality Assurance Engineer
Opentext

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE
PILANI (RAJASTHAN)
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ABSTRACT

The exponential growth of online news content has made timely information processing increasingly challenging for readers, professionals, and organisations alike. Automatic text summarisation systems that can condense lengthy news articles into concise, accurate, and fluent summaries hold significant practical value, particularly in enterprise content management and information retrieval contexts.

Existing summarisation approaches are broadly divided into two paradigms: extractive methods, which preserve factual accuracy but produce disjointed output, and abstractive methods, which leverage pre-trained transformer models such as BART and PEGASUS to generate fluent text but risk factual hallucination on longer documents. This dissertation proposes and implements a grounded, two-stage summarisation pipeline that integrates both paradigms. A hybrid evidence retrieval stage, combining BM25 lexical scoring with sentence-embedding similarity, first identifies the most salient sentences from the source article within the abstractive model's input budget. A fine-tuned BART model then generates multiple candidate summaries using nucleus sampling. Each candidate is evaluated against the retrieved evidence using a Natural Language Inference (NLI) based verifier that computes a Factual Consistency Score (FCS), and a preference reranker selects the most factually consistent and fluent candidate, with an extractive fallback mechanism when confidence falls below threshold.

At the mid-semester milestone, the dataset preparation, four baseline implementations (Lead-3, TextRank, zero-shot BART, and fine-tuned single-stage BART), and the complete evidence-retrieval, generation, verification, and reranking pipeline have been built and are undergoing systematic evaluation. Remaining work includes ablation studies, human evaluation, a multilingual feasibility study on Hindi XL-Sum, and development of a demonstration application.

Keywords: Automatic Text Summarisation, Natural Language Processing, BART, Evidence Retrieval, Factual Consistency, Natural Language Inference, BM25, Transformer Models.

P.V.Sai Charan Veeraswamy Ponnuru

Signature of the Student	Signature of the Supervisor
Name: Puthineedi Venkata Sai Charan	Name: Veeraswamy Ponnuru
Date: 20/06/2026	Date: 20/06/2026
Place: Hyderabad	Place: Hyderabad

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1. INTRODUCTION AND PROBLEM CONTEXT

The rapid proliferation of digital news content has created an information overload challenge for individuals and organisations. Manual review of news articles at scale is neither practical nor scalable. Automatic text summarisation addresses this challenge by producing condensed, coherent representations of source documents. Despite substantial research progress, existing systems face persistent limitations in factual accuracy: abstractive models generate fluent text but often introduce claims not grounded in the source, while purely extractive approaches avoid hallucination at the cost of fluency and conciseness.

This dissertation project — titled A Two-Stage Summarization Pipeline for News Articles: Extraction-Guided Abstractive Generation — addresses the factual grounding problem by designing a modular pipeline that sequences evidence retrieval, abstractive generation, NLI-based verification, and preference reranking with an extractive fallback. The proposed system targets the CNN/DailyMail benchmark and is evaluated using ROUGE, BERTScore, and NLI-based Factual Consistency Score (FCS) metrics.

1.1 Scope at the Mid-Semester Stage

By the mid-semester milestone, the initial three phases have been completed: dataset preparation and corpus analysis, baseline implementation and evaluation, and evidence retrieval, abstractive generation, NLI verification, and preference reranking pipeline development. The remaining phases cover ablation studies, human evaluation, multilingual feasibility analysis, and dissertation write-up.

- Dataset acquisition, preprocessing, and corpus analysis on CNN/DailyMail
- Four baseline systems implemented and evaluated on ROUGE, BERTScore, and FCS metrics
- Evidence retrieval module developed using hybrid BM25 and sentence-embedding scoring
- Best-of-N abstractive generation, NLI verification, and preference reranker with extractive fallback implemented
- Preliminary pipeline evaluation against baselines completed, PEGASUS and mBART comparison study in progress

2. OBJECTIVES AND RESEARCH QUESTIONS

The primary objectives of this dissertation are as follows:

- Achieve ROUGE-2 $\geq$ 18.0 on the CNN/DailyMail test set, surpassing the Lead-3 extractive baseline
- Achieve NLI-FCS $\geq$ 0.65 for the proposed pipeline, compared to $\leq$ 0.55 for single-stage fine-tuned BART
- Maintain extractive fallback rate below 15% of test documents
- Achieve human evaluation mean factual accuracy $\geq$ 4.0/5.0 with Cohen's Kappa $\geq$ 0.6

- Survey and compare extractive and abstractive summarisation methods through a structured literature review
- Conduct a multilingual feasibility study on Hindi XL-Sum comparing translate-then-summarise versus direct mBART.

2.1 Research Questions

The following research questions guide the evaluation framework:

1. Does hybrid BM25 and sentence-embedding retrieval improve factual consistency compared to truncation-based input selection?
2. Does Best-of-N generation with NLI-based reranking improve FCS over single-candidate greedy decoding?
3. Does the extractive fallback mechanism prevent low-confidence abstractive outputs from reaching the final summary?
4. How does the proposed pipeline compare to BART-large, PEGASUS, and mBART under matched conditions on CNN/DailyMail?
5. Is the translate-then-summarise approach viable for Hindi summarisation relative to direct mBART fine-tuning on XL-Sum?

3. SYSTEM MODULES COMPLETED UP TO MID-SEMESTER

3.1 Dataset Preparation and Corpus Analysis

The CNN/DailyMail benchmark dataset was acquired and split into training, validation, and test sets (70/15/15 ratio). Corpus statistics including average article length, summary length, and vocabulary coverage were computed. A smaller Hindi XL-Sum subset was also prepared for the multilingual feasibility study.

3.2 Lead-3 Baseline

Returns the first three sentences of each article, exploiting the journalistic front-loading convention. This baseline establishes a practical lower bound that any meaningful summarisation system should surpass.

3.3 TextRank Baseline

An unsupervised graph-based extractive approach in which sentences are modelled as nodes and edges are weighted by lexical similarity. Sentences are ranked by eigenvector centrality and the top-ranked sentences form the summary. Retained as both a baseline and an ablation comparator for the evidence retrieval stage.

3.4 Zero-Shot BART Baseline

A pre-trained BART-large-cnn model applied without domain-specific fine-tuning. This baseline isolates the contribution of fine-tuning on CNN/DailyMail from the architectural benefits of the pre-trained model.

3.5 Fine-Tuned Single-Stage BART Baseline

BART-large fine-tuned on CNN/DailyMail for three epochs. The model encodes a truncated article prefix and decodes a single summary without any retrieval pre-filtering or NLI verification. This baseline represents the standard single-stage abstractive pipeline against which the proposed two-stage system is compared.

3.6 Evidence Retrieval Module

Sentences from the source article are scored using a hybrid combination of BM25 lexical scoring and sentence-embedding cosine similarity. The highest-scoring sentences within the abstractive model's 1024-token budget form both the selected context for generation and the evidence pool for downstream verification.

3.7 Best-of-N Abstractive Generation Module

The fine-tuned BART model generates N = 4–5 diverse candidate summaries using nucleus sampling (top_p = 0.92), producing varied phrasings of the same content rather than a single greedy output.

3.8 NLI Evidence Verification Module

Each candidate summary is decomposed into sentences. For each sentence, the best-matching evidence sentence from the retrieved pool is identified and an entailment probability is computed using a DeBERTa-based NLI model. The mean entailment probability across all sentences in a candidate forms its Factual Consistency Score (FCS).

3.9 Preference Reranker with Extractive Fallback

Candidates are ranked using a weighted combination of FCS and BERTScore-F1. The top-ranked candidate is selected as output. If no candidate exceeds a minimum FCS threshold, the system falls back to an extractive summary derived from the retrieved evidence set, ensuring the system never silently returns an unverified output.

4. SYSTEM ARCHITECTURE AND METHODOLOGY

The proposed system follows a modular, four-stage pipeline that extends conventional two-stage extractive–abstractive summarisation with an explicit verification and selection layer. Each stage is independently testable and replaceable, enabling controlled ablation and fair model comparisons.

4.1 High-Level Architecture

The pipeline processes a source news article through four sequential stages:

- Stage 1 – Evidence Retrieval: Sentences are scored using hybrid BM25 and sentence-embedding cosine similarity. The top-scoring sentences within the 1024-token budget constitute the selected context and evidence pool.

- Stage 2 – Abstractive Generation: The fine-tuned BART model generates multiple candidate summaries via nucleus sampling from the selected context.
- Stage 3 – NLI Evidence Verification: Each candidate sentence is verified against the evidence pool using a DeBERTa-based NLI model, producing a per-candidate FCS.
- Stage 4 – Preference Reranking and Fallback: Candidates are ranked by weighted FCS and BERTScore-F1. The highest-ranked candidate is returned, or an extractive summary is provided if no candidate clears the confidence threshold.

Two-Stage Grounded Summarisation Pipeline — Architecture										
INPUT News Article	->	**Stage 1 Evidence Retrieval**	->	**Stage 2 Abstractive Generation**	->	**Stage 3 NLI Verification**	->	**Stage 4 Preference Reranking**	->	**OUTPUT Factual Summary**
		Hybrid BM25 + embedding scoring		N-candidate nucleus sampling		Sentence-level entailment score (FCS)		Weighted rank + extractive guard		

Figure 1: High-level architecture of the proposed grounded summarisation pipeline.

4.2 Background and Literature Review

The project sits at the intersection of Natural Language Processing, Information Retrieval, and Deep Learning. Key prior works informing the design include:

- TextRank (Mihalcea & Tarau, 2004): Unsupervised graph-based extractive summarisation — retained as an ablation baseline for the evidence retrieval stage.
- BART (Lewis et al., 2020): Denoising autoencoder pre-trained on span corruption, fine-tuned on CNN/DailyMail achieving approximately ROUGE-2 of 21. Serves as the primary abstractive generator in the pipeline.
- PEGASUS (Zhang et al., 2020): Gap-sentence generation pre-training, included as a domain-pretrained comparison baseline.
- mBART (Liu et al., 2020): Multilingual denoising pre-training across 25 languages supports the Hindi XL-Sum multilingual feasibility study.
- BERTScore (Zhang et al., 2020): Semantic similarity metric using contextual embeddings, used as the fluency/relevance component of the reranking score.
- SummaC (Laban et al., 2022): NLI-based inconsistency detection framework, informs the sentence-level FCS design.
- Mujahid et al. (ACL 2026): Stress-testing of factuality metrics on long-document summarisation, confirms NLI-based metrics remain reliable for CNN/DailyMail-scale documents.

5. TOOLS AND TECHNOLOGIES USED

The following components and libraries are employed across the pipeline stages:

Component	Technology / Library
Dataset & Preprocessing	HuggingFace Datasets, NLTK
Evidence Retrieval	rank_bm25, sentence-transformers (all-MiniLM-L6-v2)
Abstractive Generation	HuggingFace Transformers (BART-large, PEGASUS, mBART), PyTorch
NLI Verification	cross-encoder/nli-deberta-v3-base
Reranking & Evaluation	bert-score, rouge-score, NumPy
Demo Application	Streamlit

6. METHODOLOGY AND PIPELINE IMPLEMENTATION

6.1 Evidence Retrieval Module

Each sentence in the source article is scored using a hybrid of BM25 lexical scoring and sentence-embedding cosine similarity against the document's overall information density. The two scores are linearly combined. The top-K sentences within the abstractive model's 1024-token input budget are selected, forming both the generation context and the verification evidence pool. TextRank is retained as an ablation comparator to quantify the benefit of the hybrid approach over graph-based salience ranking.

6.2 Abstractive Generation

BART-large is fine-tuned on the CNN/DailyMail training split for three epochs with a learning rate of 3×10^{-5} and a batch size of 2 with gradient accumulation of 8 steps. At inference time, N = 4–5 diverse candidate summaries are generated using nucleus sampling (top_p = 0.92). This Best-of-N approach broadens the hypothesis space, providing the downstream verification and reranking stages with meaningful alternatives to discriminate between.

6.3 NLI Evidence Verification

For each candidate summary, individual sentences are verified against the evidence pool using a compact NLI model. Each summary sentence is matched to its best-supporting evidence sentence, and an entailment probability is computed. The mean entailment probability across all sentences in a candidate constitutes its Factual Consistency Score (FCS). This sentence-level approach provides a tractable approximation of full atomic-claim verification without requiring additional claim-decomposition infrastructure.

6.4 Preference Reranking and Extractive Fallback

Candidates are ranked using a weighted score:

```
final_score = 0.6 × FCS + 0.4 × BERTScore_F1
```

The candidate with the highest combined score is selected. If even the top-ranked candidate falls below a minimum FCS threshold, the system falls back to the extractive summary produced directly from the retrieved evidence set. The fallback activation rate is tracked as a system-health metric.

7. EXPERIMENTAL SETUP AND BASELINES

All systems are evaluated on the same CNN/DailyMail test split (15% of the full dataset). Automatic evaluation uses ROUGE-1, ROUGE-2, ROUGE-L, BERTScore-F1, and NLI-FCS. To rigorously assess the proposed pipeline, the following baseline models have been implemented for comparative analysis:

- Lead-3 Baseline: A standard extractive heuristic that selects the first three sentences of an article, serving as the fundamental benchmark for positional bias.
- TextRank Baseline: An unsupervised, graph-based extractive approach that ranks sentences by establishing semantic connectivity within the document.
- Zero-Shot BART Baseline: A pre-trained transformer model used to gauge the base abstractive generative capabilities without task-specific fine-tuning.
- Fine-Tuned BART Baseline: A standard, single-stage end-to-end abstractive model representing the conventional approach, trained specifically on the CNN/DailyMail corpus.
- Evaluation Metrics: ROUGE evaluates lexical overlap, BERTScore measures semantic equivalence, and NLI-FCS specifically penalises factual inconsistencies and hallucinations.

8. CURRENT IMPLEMENTATION STATUS

Work Package	Deliverable	Status	Evidence Expected
Dataset preparation	CNN/DM splits, corpus statistics	COMPLETED	Corpus statistics report
Baseline implementation	Lead-3, TextRank, zero-shot BART, fine-tuned BART	COMPLETED	ROUGE, BERTScore, FCS tables
Evidence retrieval module	Hybrid BM25 + embedding scorer	COMPLETED	Ablation retrieval results
Abstractive generation	Fine-tuned BART, Best-of-N sampling	COMPLETED	Generation outputs
NLI verification & reranker	FCS computation, fallback logic	COMPLETED	FCS distribution analysis
PEGASUS/mBART comparison	Comparative evaluation	IN PROGRESS	Final metric tables

Work Package	Deliverable	Status	Evidence Expected
Ablation studies	Component-wise ablation	PENDING	Ablation result tables
Human evaluation	50 samples, 3 raters, Likert scale	PENDING	Inter-rater agreement, Kappa
Demo application	Streamlit pipeline wrapper	PENDING	Working demo

9. TECHNICAL SPECIFICATIONS OF THE PROPOSED SYSTEM

Sl. No.	Technical Parameter	Proposed Specification
1	Primary dataset	CNN/DailyMail (70/15/15 split)
2	Secondary dataset	Hindi XL-Sum subset (multilingual feasibility)
3	Evidence retrieval	Hybrid BM25 + all-MiniLM-L6-v2 sentence embeddings (384-dim)
4	Primary generator	BART-large-cnn, fine-tuned 3 epochs, lr=3e-5, batch 2, grad accum 8
5	Comparison generators	PEGASUS-cnn_dailymail, mBART-large-cc25
6	Generation strategy	Nucleus sampling, top_p = 0.92, N = 4–5 candidates
7	NLI verifier	cross-encoder/nli-deberta-v3-base (sentence-level entailment)
8	Reranking formula	0.6 × FCS + 0.4 × BERTScore-F1
9	Evaluation metrics	ROUGE-1/2/L, BERTScore-F1, NLI-FCS, Cohen's Kappa
10	Human evaluation	50 test samples, 3 raters, 1–5 Likert scale (fluency, factual accuracy, coherence)
11	Demo application	Streamlit (local), displays summary, evidence trace, FCS, fallback status

10. DESIGN CONSIDERATIONS

The following principles guide the architectural and implementation choices throughout the project:

- Modularity: Each stage retrieval, generation, verification, and reranking is designed as an independently testable and replaceable module, enabling controlled ablation and straightforward extension.
- Traceability: Every generated summary is accompanied by an evidence trace linking it to specific source sentences, supporting verification and manual review.
- Graceful degradation: The extractive fallback ensures the system never silently returns a low-confidence or potentially hallucinated abstractive summary.
- Evaluation rigour: The pipeline evaluation harness accepts BART, PEGASUS, or mBART under identical conditions, enabling fair like-for-like comparison across generator variants.
- Reproducibility: All model versions, hyperparameters, and random seeds are documented, dataset splits are fixed and versioned for consistent replication.
- Scope management: Production deployment and browser integration are explicitly out of scope; a local Streamlit demonstration satisfies presentability requirements without scope creep.

11. PRELIMINARY OBSERVATIONS AND RISKS

11.1 Preliminary Observations

- Hybrid BM25 and embedding retrieval consistently selects semantically coherent evidence sentences that are not always captured by pure keyword overlap, suggesting the dual-signal approach is necessary for diverse article styles.
- Best-of-N sampling with nucleus decoding (top_p = 0.92) produces candidates with meaningfully different phrasings, providing non-trivial alternatives for the downstream reranker to discriminate between.
- Preliminary NLI scoring reveals that higher-FCS candidates tend to use more conservative phrasing and shorter sentences, indicating a tension between fluency and verifiability that the weighted reranking formula must balance.
- The extractive fallback is activated for a small proportion of test documents where abstractive candidates collectively fall below the confidence threshold, suggesting the fallback is a genuine safety net rather than a dominant mode.
- PEGASUS's shorter 512-token encoder limit compared to BART's 1024-token limit interacts with the retrieval stage, where evidence sentences sometimes exceed the PEGASUS budget, warranting separate ablation treatment.

11.2 Current Risks and Mitigations

Risk	Mitigation
NLI metric reliability on longer documents	Restrict primary evaluation to CNN/DailyMail (short-document setting), cite Mujahid et al. (2026) as justification for excluding long-document benchmarks.
Fine-tuning compute cost and time overrun	Colab Pro+ and Kaggle A100 GPU allocation gradient accumulation reduces memory pressure checkpoint resumption if session interrupts.

Risk	Mitigation
Human evaluation annotator availability	Annotator pool identified in advance evaluation limited to 50 samples across 3 raters to control effort and scheduling.
Multilingual quality degradation under translate-then-summarise	Direct mBART fine-tuning on XL-Sum provides a fallback path study that is framed as exploratory feasibility rather than a primary contribution.
Metric sensitivity to FCS threshold selection	Threshold is treated as an ablation variable; results are reported across a range of threshold values to demonstrate robustness.

12. FUTURE PLAN

The plan below follows the approved dissertation schedule. Phases 1 through 3 are considered completed for this mid-semester report. The remaining phases focus on evaluation, refinement, review, and final submission.

Sl. No.	Phase	Start Date – End Date	Work to be Done	Status
1	Dissertation Outline & Setup	25 Apr 2026 – 10 May 2026	Literature review, project environment setup, dataset identification	COMPLETED
2	Design & Baseline Development	11 May 2026 – 31 May 2026	Pipeline design, baseline implementation, dataset preprocessing	COMPLETED
3	Pipeline Development & Mid-Semester Report	01 Jun 2026 – 21 Jun 2026	Evidence retrieval, generation, verification, reranking modules, mid-semester report compilation	COMPLETED
4	Testing & Evaluation	22 Jun 2026 – 12 Jul 2026	Ablation studies, human evaluation, multilingual feasibility study, error analysis, attention visualisation, demo application	PENDING
5	Final Review & Submission	13 Jul 2026 – 02 Aug 2026	Supervisor and examiner review, final dissertation chapters, presentation preparation, VIVA portal submission	PENDING

13. ABBREVIATIONS

Abbreviation	Full Form
NLP	Natural Language Processing
BART	Bidirectional and Auto-Regressive Transformer
PEGASUS	Pre-training with Extracted Gap-Sentences for Abstractive SUmmarization
mBART	Multilingual Bidirectional and Auto-Regressive Transformer
ROUGE	Recall-Oriented Understudy for Gisting Evaluation
BERTScore	BERT-based Semantic Similarity Metric for Text Generation
NLI	Natural Language Inference
FCS	Factual Consistency Score
BM25	Best Matching 25 (Probabilistic Lexical Retrieval Function)
CNN/DM	CNN/DailyMail Summarisation Benchmark Dataset
XL-Sum	Cross-Lingual Summarisation Dataset
RAG	Retrieval-Augmented Generation
NER	Named Entity Recognition
GPU	Graphics Processing Unit
API	Application Programming Interface

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